

PRILL'S PROBLEM IN EVERY GENUS $g \geq 2$

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ABSTRACT. Let Y be a smooth projective connected complex curve of genus at least two. We construct a connected finite étale Galois cover $f : X \rightarrow Y$ for which every fiber divisor $f^{-1}(y)$ moves in a pencil. The essential input is Raynaud's bundle P_g , obtained by restricting the Fourier–Mukai transform of $\mathcal{O}_{\mathrm{Pic}^0(Y)}(-g\Theta)$, together with Mukai's formula showing that this transform splits as copies of one line bundle after a polarization isogeny. We choose a degree -1 twist with torsion determinant, which turns that splitting into a finite étale trivialization; the regular representation of the resulting deck group then supplies the required second section.

1. INTRODUCTION

All curves and varieties are defined over $\mathbb{C}$. Let $f : X \rightarrow Y$ be a finite morphism of smooth projective connected curves, and write $D_y = f^*y$ for its scheme-theoretic fiber over $y \in Y$. Prill asked whether

$$h^0(X, \mathcal{O}_X(D_y)) = 1$$

for a general point y whenever $g(Y) \geq 2$; see [ACGH85, Chapter VI, Exercise D]. The condition $h^0(X, \mathcal{O}_X(D_y)) \geq 2$ is closed in y by upper semicontinuity. Hence, for a fixed cover, it holds for a general point if and only if it holds for every point. Following Landesman and Litt, such a cover is called *Prill exceptional*. They constructed a finite étale Prill-exceptional cover of every complex curve of genus two and asked whether examples exist in higher genus [LL24, Theorem 1.2 and Remark 1.4].

We answer this question in every genus, and the cover we obtain is Galois.

Theorem 1.1. *Let Y be a smooth projective connected complex curve of genus $g \geq 2$. There is a connected finite étale Galois cover $f : X \rightarrow Y$ such that*

$$h^0(X, \mathcal{O}_X(f^{-1}(y))) \geq 2 \quad \text{for every } y \in Y.$$

The proof begins with Raynaud's bundle P_g on Y . This bundle has two complementary properties. First, it has no theta divisor:

$$H^0(Y, P_g \otimes L) \neq 0 \quad \text{for every } L \in \mathrm{Pic}^0(Y).$$

Second, Mukai's pullback formula says that the vector bundle on the Jacobian from which P_g is restricted becomes a direct sum of copies of a single line bundle after a finite isogeny. Thus, after a finite étale base change, the associated projective bundle is already trivial; the only remaining obstruction to trivializing the vector bundle is one scalar line bundle.

The choice of the index g is essential. Raynaud's bundle P_m has slope g/m , so P_g has slope one. We can therefore choose a line bundle B of degree -1 such that

$$E = P_g \otimes B, \quad H^0(Y, E) = 0, \quad \det E \in \mathrm{Pic}^0(Y)_{\mathrm{tors}},$$

while the Raynaud nonvanishing still gives $H^0(Y, E(y)) \neq 0$ for every y . Mukai's formula makes a finite étale pullback of E equal to $M^{\oplus g}$ for one line bundle M . The torsion determinant forces M itself to be torsion, so a further finite étale cover makes E trivial. The bundle E then comes from a representation of a finite group. One irreducible constituent has a section after twisting by every point of Y , and the same constituent occurs in the regular representation, hence in the pushforward of the structure sheaf of the final cover.

2. THE RAYNAUD–MUKAI MECHANISM

Fix a point $y_0 \in Y$, put $A = \mathrm{Pic}^0(Y)$, and let $J = \widehat{A}$ be the dual abelian variety. Write $\iota : Y \hookrightarrow J$ for the Abel–Jacobi embedding determined by y_0 . We use the theta divisor attached to the same base point,

$$\Theta = \{\alpha \in \mathrm{Pic}^0(Y) : H^1(Y, \alpha((g-1)y_0)) \neq 0\},$$

as in Hein's normalization of Raynaud's construction [Hei07, §2.1]. Let $\mathcal{P}$ be the Poincaré bundle on $J \times A$, normalized along $\{0\} \times A$ and $J \times \{0\}$, and let $p : J \times A \rightarrow A$ and $q : J \times A \rightarrow J$ be the projections.

The Fourier–Mukai transform with kernel $\mathcal{P}$ is

$$\Phi_{\mathcal{P}}(F) = Rq_*(\mathcal{P} \otimes p^*F), \quad F \in D^b(A).$$

For $\xi \in J = \text{Pic}^0(A)$, write $\mathcal{P}_{\xi} = \mathcal{P}|_{\{\xi\} \times A}$. Proper base change identifies the derived fiber at ξ with

$$L\xi^* \Phi_{\mathcal{P}}(F) \simeq R\Gamma(A, F \otimes \mathcal{P}_{\xi}).$$

Thus the transform packages, in one object on J , the cohomology of all degree-zero twists of F . Raynaud applies it to the inverse of a positive multiple of the theta line bundle.

For an ample line bundle $\mathcal{N}$ on A , the homomorphism

$$\varphi_{\mathcal{N}} : A \longrightarrow \widehat{A}, \quad x \longmapsto t_x^* \mathcal{N} \otimes \mathcal{N}^{-1},$$

measures the failure of $\mathcal{N}$ to be invariant under translation. It is an isogeny. Mukai’s formula says that this same isogeny completely splits the Fourier–Mukai transform of $\mathcal{N}^{-1}$.

Proposition 2.1 (Mukai–Raynaud). *For $m \geq 1$, set $\mathcal{N}_m = \mathcal{O}_A(m\Theta)$ and*

$$\widetilde{\mathcal{R}}_m = R^g q_*(\mathcal{P} \otimes p^* \mathcal{N}_m^{-1}), \quad \mathcal{R}_m = [-1_J]^* \widetilde{\mathcal{R}}_m, \quad P_m = \iota^* \mathcal{R}_m.$$

Then P_m is semistable and

$$\text{rk}(P_m) = m^g, \quad \deg(P_m) = gm^{g-1}, \quad H^0(Y, P_m \otimes L) \neq 0 \quad \text{for every } L \in \text{Pic}^0(Y).$$

Let $\varphi_m = \varphi_{\mathcal{N}_m} : A \rightarrow J$, and put $a_m = [-1_J] \circ \varphi_m$. Then

$$a_m^* \mathcal{R}_m \simeq \mathcal{N}_m^{\oplus m^g}.$$

Proof. For $\xi \in J$, the line bundle $\mathcal{N}_m \otimes \mathcal{P}_{\xi}^{-1}$ is ample. Since $\omega_A \simeq \mathcal{O}_A$, Serre duality and Kodaira vanishing give

$$H^i(A, \mathcal{N}_m^{-1} \otimes \mathcal{P}_{\xi}) = 0 \quad (i \neq g).$$

Riemann–Roch on the principally polarized abelian variety A gives

$$h^g(A, \mathcal{N}_m^{-1} \otimes \mathcal{P}_{\xi}) = h^0(A, \mathcal{N}_m \otimes \mathcal{P}_{\xi}^{-1}) = \chi(\mathcal{N}_m) = m^g.$$

Cohomology and base change therefore show that $\Phi_{\mathcal{P}}(\mathcal{N}_m^{-1})$ is concentrated in degree g , and that $\widetilde{\mathcal{R}}_m$ is a vector bundle of rank m^g .

Mukai’s pullback formula, applied to $\mathcal{N}_m$, is

$$\varphi_m^* \widetilde{\mathcal{R}}_m \simeq \mathcal{N}_m^{\oplus h^0(A, \mathcal{N}_m)} = \mathcal{N}_m^{\oplus m^g};$$

see [Muk81, Proposition 3.11(1)]. Raynaud records and uses exactly this formula in [Ray82, §3.1]. Since $a_m = [-1_J] \circ \varphi_m$ and $\mathcal{R}_m = [-1_J]^* \widetilde{\mathcal{R}}_m$, the displayed formula is equivalent to $a_m^* \mathcal{R}_m \simeq \mathcal{N}_m^{\oplus m^g}$.

Restricting this identity to the inverse image of the Abel–Jacobi curve shows why the resulting bundle is semistable: after a finite étale base change it is a direct sum of copies of one line bundle. Raynaud’s intersection calculation on that inverse-image curve gives $\mu(P_m) = g/m$, and hence $\deg(P_m) = gm^{g-1}$. Hein records the same construction, including the pullback by $[-1_J]$, and the resulting rank, degree, slope, and semistability in [Hei07, §2, Lemma 2.3].

It remains to explain the nonvanishing. The second Fourier–Mukai identity used by Raynaud identifies the transform of $\widetilde{\mathcal{R}}_m$ back on A with the line bundle $(-1_A)^* \mathcal{N}_m^{-1}$. Using this identity, Raynaud produces a nonzero family of sections after restricting to an Abel–Jacobi translate, and hence obtains $H^0(Y, P_m \otimes L) \neq 0$ for general $L \in \text{Pic}^0(Y)$; see [Ray82, §3.1]. Hein’s definition above is the corresponding normalized form of Raynaud’s bundle. The locus

$$\{L \in \text{Pic}^0(Y) : H^0(Y, P_m \otimes L) \neq 0\}$$

is closed by upper semicontinuity. Since it contains a dense open subset, it is all of $\text{Pic}^0(Y)$. This all-twists formulation is also recorded explicitly in [Hei07, Introduction]. $\square$

Remark 2.2 (Why the determinant is made torsion). Let $r = m^g$, and consider a twist $E = P_m \otimes B$. Pulling the isogeny a_m back along the Abel–Jacobi embedding gives a finite étale cover $\rho : Z \rightarrow Y$. Mukai’s formula then takes the form

$$\rho^* E \simeq M^{\oplus r}$$

for a line bundle M on Z . Hence

$$\mathbb{P}(\rho^* E) \simeq Z \times \mathbb{P}^{r-1}, \quad M^{\otimes r} \simeq \det(\rho^* E) = \rho^*(\det E).$$

The first identity says that the projective part of the bundle has disappeared, but it does not yet trivialize the vector bundle: the common factor M is the remaining obstruction. The determinant detects exactly

this factor. If $\det E$ is torsion, then the second identity shows that $M^{\otimes r}$, and hence M , is torsion. A finite étale cover associated with a trivialization of a power of M then makes M , and therefore E , trivial. Conversely, if a finite cover $u : U \rightarrow Z$ trivializes M , then

$$\mathrm{Nm}_u(u^*M) \simeq M^{\otimes \deg u}$$

is trivial, so M must already be torsion. Thus the torsion-determinant condition is precisely what turns Mukai's projective trivialization into a genuine finite étale trivialization of the vector bundle.

3. A DEGREE-ZERO BUNDLE WITH FINITE ÉTALE TRIVIALIZATION

We now take $m = g$ and put $r = g^g$. Proposition 2.1 gives

$$\mathrm{rk}(P_g) = \deg(P_g) = r,$$

so P_g has slope one.

Lemma 3.1. *If F is a semistable vector bundle of degree zero on Y , then*

$$V^0(F) = \{L \in \mathrm{Pic}^0(Y) : H^0(Y, F \otimes L) \neq 0\}$$

is finite.

Proof. Choose a Jordan–Hölder filtration $0 = F_0 \subset F_1 \subset \cdots \subset F_N = F$, with stable degree-zero quotients $Q_j = F_j/F_{j-1}$. If $L \in V^0(F)$, a nonzero section of $F \otimes L$ gives a nonzero map $L^{-1} \rightarrow F$. The saturation of its image is $L^{-1}(D)$ for an effective divisor D . Semistability of F gives $\deg D \leq 0$, hence $D = 0$; thus L^{-1} is a degree-zero line subbundle of F .

Choose the least j for which $L^{-1} \subset F_j$. The induced map $L^{-1} \rightarrow Q_j$ is nonzero. A nonzero morphism between stable bundles of the same slope is an isomorphism. Hence $Q_j \simeq L^{-1}$, and in particular Q_j has rank one. Thus L^{-1} is isomorphic to one of the finitely many rank-one Jordan–Hölder factors of F . $\square$

Proposition 3.2. *There is a vector bundle E on Y such that*

$$\deg E = 0, \quad H^0(Y, E) = 0, \quad \det E \in \mathrm{Pic}^0(Y)_{\mathrm{tors}}, \quad H^0(Y, E(y)) \neq 0 \quad \text{for every } y \in Y.$$

Proof. Choose $B_0 \in \mathrm{Pic}^{-1}(Y)$ and set $E_0 = P_g \otimes B_0$. Since P_g is semistable of slope one, E_0 is semistable of degree zero. Let $D = \det(E_0) \in \mathrm{Pic}^0(Y)$. Multiplication by r on the abelian variety $\mathrm{Pic}^0(Y)$ is surjective, so choose $L_0 \in \mathrm{Pic}^0(Y)$ with $L_0^{\otimes r} \simeq D^{-1}$.

The torsion subgroup of $\mathrm{Pic}^0(Y)$ is infinite, whereas $V^0(E_0)$ is finite by Lemma 3.1. We may therefore choose a torsion line bundle η such that $L_0 \otimes \eta \notin V^0(E_0)$. Set

$$B = B_0 \otimes L_0 \otimes \eta, \quad E = P_g \otimes B = E_0 \otimes L_0 \otimes \eta.$$

Then $\deg E = 0$, $H^0(Y, E) = 0$, and

$$\det E \simeq D \otimes (L_0 \otimes \eta)^{\otimes r} \simeq \eta^{\otimes r},$$

so $\det E$ is torsion. For every $y \in Y$, the line bundle $B(y) = B \otimes \mathcal{O}_Y(y)$ has degree zero. Proposition 2.1 therefore gives

$$H^0(Y, E(y)) = H^0(Y, P_g \otimes B(y)) \neq 0.$$

$\square$

Proposition 3.3. *The bundle E of Proposition 3.2 is trivialized by a connected finite étale Galois cover of Y .*

Proof. Mukai's formula has already trivialized the projective bundle associated with E ; we now remove the common line-bundle factor described in Remark 2.2. Let $a = a_g : A \rightarrow J$ and $\mathcal{N} = \mathcal{N}_g$ be as in Proposition 2.1, and form the Cartesian square

$$\begin{array}{ccc} Z & \xrightarrow{\tau} & A \\ \rho \downarrow & & \downarrow a \\ Y & \xrightarrow{\iota} & J. \end{array}$$

The morphism ρ is finite étale. Choose a connected component Z_0 of Z , and write $\rho_0 : Z_0 \rightarrow Y$ and $\tau_0 : Z_0 \rightarrow A$ for the restrictions. The image of Z_0 in Y is nonempty, open, and closed, hence ρ_0 is surjective. Proposition 2.1 gives

$$\rho_0^*E \simeq (\tau_0^*\mathcal{N} \otimes \rho_0^*B)^{\oplus r}.$$

Put $M = \tau_0^*\mathcal{N} \otimes \rho_0^*B$. Taking determinants yields

$$M^{\otimes r} \simeq \rho_0^*(\det E).$$

If $(\det E)^{\otimes d} \simeq \mathcal{O}_Y$, then $M^{\otimes rd} \simeq \mathcal{O}_{Z_0}$. Thus M is torsion.

Choose $n \geq 1$ and an isomorphism $M^{\otimes n} \simeq \mathcal{O}_{Z_0}$. Using this isomorphism to define multiplication, form the $\mathcal{O}_{Z_0}$ -algebra

$$\mathcal{A} = \mathcal{O}_{Z_0} \oplus M^{-1} \oplus \dots \oplus M^{-(n-1)}$$

and the finite cover $u : U = \operatorname{Spec}_{Z_0}(\mathcal{A}) \rightarrow Z_0$. Locally this cover has equation $t^n = v$, where v is a unit. Since the characteristic is zero, the derivative nt^{n-1} is a unit on the cover, so u is étale. The tautological root trivializes u^*M , and hence $(\rho_0 \circ u)^*E \simeq \mathcal{O}_U^{\oplus r}$.

A connected component U_0 of U still maps surjectively and finite étale to Y . Let $T \rightarrow Y$ be the normalization of Y in a Galois closure of the finite extension $\mathbb{C}(U_0)/\mathbb{C}(Y)$. Conjugates of an unramified extension of function fields are unramified, and so is their compositum. Hence $T \rightarrow Y$ is a connected finite étale Galois cover dominating $U_0 \rightarrow Y$, and the pullback of E to T is trivial. $\square$

4. THE GALOIS COVER

We fix the descent convention used below. Let $q : X \rightarrow Y$ be a connected finite étale Galois cover with group G , acting on X on the right. For a finite-dimensional left G -module V , set

$$\mathcal{E}(V) = X \times^G V = (X \times V)/((xg, v) \sim (x, gv)).$$

Lemma 4.1. *With this convention:*

- (1) *every vector bundle on Y that becomes trivial on X is isomorphic to $\mathcal{E}(V)$ for a finite-dimensional G -module V ;*
- (2) $H^0(Y, \mathcal{E}(V)) \simeq V^G$;
- (3) $q_*\mathcal{O}_X \simeq \mathcal{E}(\operatorname{Fun}(G, \mathbb{C}))$, where $(g \cdot \varphi)(h) = \varphi(g^{-1}h)$.

Proof. Choose a trivialization $q^*F \simeq \mathcal{O}_X \otimes V$ of a vector bundle F . In this trivialization its descent isomorphisms are regular maps $X \rightarrow \operatorname{GL}(V)$. Every such map is constant because X is connected and projective whereas $\operatorname{GL}(V)$ is affine. The cocycle identity therefore gives a linear action of G on V , and the descended bundle is $\mathcal{E}(V)$.

A section of $\mathcal{E}(V)$ pulls back to a regular map $\phi : X \rightarrow V$ satisfying $\phi(xg) = g^{-1}\phi(x)$. The map ϕ is constant because X is projective and V is affine, and the equivariance condition says that its value lies in V^G . This proves the second assertion.

Étale-locally on Y , a section of the G -torsor identifies X with $Y \times G$ and $q_*\mathcal{O}_X$ with $\mathcal{O}_Y \otimes \operatorname{Fun}(G, \mathbb{C})$. Changing the local section by right multiplication by g^{-1} changes a function φ to $h \mapsto \varphi(g^{-1}h)$, which gives the stated associated bundle. $\square$

Proof of Theorem 1.1. Let E be the bundle constructed in Proposition 3.2, and choose a connected finite étale Galois cover $f : X \rightarrow Y$ that trivializes it, as supplied by Proposition 3.3. Let G be the deck group. By Lemma 4.1, $E \simeq \mathcal{E}(V)$ for a finite-dimensional complex representation V of G . Maschke's theorem gives

$$V \simeq \bigoplus_{i=1}^s V_i^{\oplus n_i}, \quad E \simeq \bigoplus_{i=1}^s W_i^{\oplus n_i}, \quad W_i = \mathcal{E}(V_i),$$

where the V_i are pairwise nonisomorphic irreducible representations. Since $H^0(Y, E) = 0$, Lemma 4.1 gives $V^G = 0$, so none of the V_i is trivial.

For each i , let

$$S_i = \{y \in Y : H^0(Y, W_i(y)) \neq 0\}.$$

This set is closed by upper semicontinuity, applied to $\operatorname{pr}_1^* W_i \otimes \mathcal{O}_{Y \times Y}(\Delta)$ over the second projection. Since $H^0(Y, E(y)) \neq 0$ for every y , the finitely many sets S_i cover Y . The curve Y is irreducible, so one of them equals Y . Fix such an index and write $W = W_i$. Then

$$H^0(Y, W(y)) \neq 0 \quad \text{for every } y \in Y.$$

By Lemma 4.1, $f_*\mathcal{O}_X \simeq \mathcal{E}(\operatorname{Fun}(G, \mathbb{C}))$. The regular representation $\operatorname{Fun}(G, \mathbb{C})$ contains both the trivial representation and V_i as direct summands. Hence $f_*\mathcal{O}_X$ contains $\mathcal{O}_Y \oplus W$ as a direct summand. Finally, $\mathcal{O}_X(f^{-1}(y)) \simeq f^*\mathcal{O}_Y(y)$, so the projection formula gives

$$H^0(X, \mathcal{O}_X(f^{-1}(y))) \simeq H^0(Y, f_*\mathcal{O}_X \otimes \mathcal{O}_Y(y)).$$

The direct summand $\mathcal{O}_Y \oplus W$ therefore gives an injection

$$H^0(Y, \mathcal{O}_Y(y)) \oplus H^0(Y, W(y)) \hookrightarrow H^0(X, \mathcal{O}_X(f^{-1}(y))).$$

The first summand contains the canonical section of the effective divisor y , and the second is nonzero by construction. Thus the target has dimension at least two for every $y \in Y$. $\square$

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